FEASIBILITY STUDY FOR CS4433 PROJECT: MODERATED LMS		
TYPE	ASSESSMENT	VERDICT
Technical	C++17, CMake and SQLite are free and run offline on any computer the school can afford. SQLite is embedded and handles 800 students comfortably with indexed queries, so a quick login or page response(say 3 seconds) is realistic. Though, SQLite files on a shared network drive can handle simultaneous writers difficultly, so the database will be in one HOST computer in the network.	Feasible
Operational	Role-based user interfaces show each user only what *they* need from the system, which supports the need for a simple system. Manuals per role help users(especially non-IT teachers)adopt it fairly smoothly. The workflow follows the roles the school already has, so little retraining is needed beyond the manuals and maybe a short demo. The system administrator is the natural local support after handover(I will handle what NEEDS me).	Feasible
Economic	All tools used are free. No new hardware or internet is needed. The main cost is the resource(ME). The benefits far outweigh that cost(ME!), observe: less manual marking for teachers, controlled release by HoD and a full audit trail of mark changes. If the system is not installed: manual marking(takes time, more room for error due to fatigue), Spreadsheets have no role control or audit log. A small custom system fits the specific constraints(offline,old-hardware)the best,compared to already existing LMSs like	Feasible

	Moodle, TalentLMS or Docebo. Effort is roughly 37 days of my own time(Cheap)	
Legal	The system holds learners' and the staff's personal data and assessment records, so it must comply with Lesotho's data privacy legislation(Data protection act,2012). Design response: salted password hashes, role-based access,audit log, store only data that is needed. The learners are minors, so guardian/parental consent and extra care with their data apply. SQLite is pblic and GoogleTest is BSD-licensed, so there are no fees or restrictions(legally and otherwise)	Feasible if proper steps of the law are followed
Schedule	Planned to take about 37 days divided in weeks with fixed milestones. Time is the highest exposure risk, so this is achievable if the scope is controlled, steady weekly hours and certain features can be dropped. With buffer days in the last week	Constricted but feasible
Recommendation: GO, with strict conditions. The system is feasible on all five dimensions. Time and legal compliance are the main constraints, addressed through scope control and the security design. The strict conditions are: 1. The database must be in one computer(host), 2. MUST-HAVE requirements are built before COULD-HAVE requirements and 3. Data protection is PROPERLY designed from the start.		

RISK REGISTER

Likelihood(1-5): 1. Rare
2. Unlikely
3. Possible
4. Likely
5. Almost certain

Impact(1-5): 1. Negligible
2. Minor
3. Moderate
4. Major
5. Severe

Exposure bands: 1-4: low
5-9: medium
10-14: high
15-19:very high
20-25: Critical

Risk	Type	Likelihood	Impact	Exposure	Mitigation	owner
Time running out	Project	4	5	20	Frequently check the Gantt Chart; Drop the COULD-HAVE features first	me
Stakeholder conflicts resolved wrongly	Project	3	4	12	Properly document each resolution and reason and confirm with the supervisor at milestone 1	me
Laptop failure or lost work	Project	2	5	10	Commit to GitHub frequently. Store every document needed in several devices	me
Illness, fatigue, power cuts,	People	3	3	9	Set a proper rest schedule. pay more attention to my physical and mental condition. keep buffer in plan	me
Password-hashing library may be hard to integrate	Technical	2	3	6	Choose a library during design; keep a fallback. Test seperately first	me

Supervisor’s availability at milestone reviews	People	2	3	6	Choose an “in-case’’ person. Let the supervisor know, when progress gets to a certain point, that I will submit for a review	me
Database schema drifts from Class Diagram as implementation progresses	Technical	3	4	12	Consistency check at every milestone.	me

STAKEHOLDER ANALYSIS AND CONFLICT RESOLUTION

Stakeholder	Main interests
Principal	Simple system for non-IT teachers; students see no marks until HoD has given the green light
Student representative	Resubmission before deadline is possible; written feedback; late submissions are possible
IT officer	Works offline on old computers; protected passwords; can handle 800 students comfortably
Head of Department	See and moderate every mark in the department before release; give late penalty of 10% mark deduction per day for 3 days, beyond 3 days the deduction is 100%
Administrator	Manage users, departments, courses, lecturer assignment and enrolment
Lecturer	Students as soon as marking is complete; ability to grant extensions where necessarry

Conflict	Resolution
Principal and HoD: withhold marks until moderated VS lecturer: students should see their marks as soon as marking is completed	Students see marks only after HoD release. The lecturer marks and the marks enter the HoD's moderation queue. This is an institutional policy and an assessment-integrity requirement
Lecturer: allow extensions VS HoD: fixed late submission penalty	The late submission penalty starts from the new extended deadline. Each extension should be properly justified
Student representative: late submissions not rejected VS HoD: 100% deduction after 3 days	The submission is accepted and marked, then the late submission penalty is applied, meaning the 0 after 3 days is the student's mark for that assessment
HoD: see every mark in the department before release VS IT officer: The system should be fast	The HoD going through every single mark individually may take too long and cause delays in the system and workflow. So in this case, HoD having the authority and access is what matters more than HoD actually using the authority and access.

Ambiguities settled:

A day late is the 24 hour period FROM the deadline. Day 1= -10% Day 2= -20% Day 3= -30% beyond Day 3= -100%

The penalty is calculated from the maximum mark possible, not the student's mark

Resubmission is allowed ONLY BEFORE the deadline, any submission beyond the deadline is the only one allowed

Written feedback is required before the HoD releases the marks, preferably even before the marks are sent for moderation